

Nathan Huse

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Full-stack developer who ships production systems and the guardrails that keep them honest: policy-gated hardware writes, access control enforced by the database rather than only the UI, and test suites that fail before output can drift.

EXPERIENCE

Contract Developer, Solid Wood Worx

2026

- Pitched the pilot and shipped it to beta on the shop floor, turning each order into a printable build sheet instead of a builder working out parts, counts and box routing by hand
- Reconciled the shop's build rules on the floor as 48 numbered questions, then validated the tool by generating a sheet that matched the checklist maintainer's hand-built sheet line for line and exposed two catalog defects
- Built the engine as a pure interpreter over versioned YAML catalogs, so every vehicle-specific fact lives in data and a new kit is a new file with zero code
- 453-test pytest suite as of 2026-07-25, covering the catalog gates, the PDF and OCR intake parser, order resolution, and the golden-locked sheet renderer

Food Service Partner, Scripps Health, Mercy Hospital San Diego

Jul 2021 - Mar 2026

- Identified ordering bottlenecks during solo operations and built a full-stack ordering platform, deployed into production use by hospital staff
- 2025 Values in Action Honoree, Scripps Mercy Hospital San Diego

PROJECTS

Grow Tent Telemetry, Sole Developer

2026 - present

- Hardware writes sit behind an explicit policy gate with an enable flag, an allowed-sources list, a per-minute rate limit, a runtime kill switch, and a protected-ports list that rejects any write to the outlet powering the Pi itself
- Reads the controller and power strip directly over Bluetooth LE by decoding the device's self-streaming packet format, with the vendor cloud reader removed from the codebase entirely
- A Raspberry Pi at the tent writes a remote Postgres system of record across a tailnet through a durable JSONL spool that buffers during network outages and backfills on recovery

CafeNightClub, Sole Developer

2025 - 2026

- Built and shipped a production ordering system for 80+ night-shift hospital staff, with real-time order tracking over Supabase Realtime WebSockets
- Order creation runs as a single database RPC call, so there are no partial writes and no race between cart validation and inventory checks
- Row-Level Security policies on every table, with role-based access (user, admin, owner) enforced server-side rather than only in the UI, and rate limiting on cart operations

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS

Frontend: React, Next.js, Astro, Tailwind CSS, DaisyUI, SSR, Responsive Design, Accessibility

Backend: Node.js, Bun, PostgreSQL, SQLite, Drizzle ORM, Supabase, better-auth, RESTful APIs, SSE, Linux (Ubuntu), Caddy, PM2

Tools & Practices: Git, GitHub, CI/CD (GitHub Actions), Tailscale, Vitest, pytest, AI-accelerated development (Claude), State Management, Database Design

AI & Search: Semantic search, Vector embeddings and reranking (Voyage AI), pgvector (HNSW), MCP tool servers, Multi-agent orchestration, Context engineering

EDUCATION

San Diego State University, Bachelor of Science in Computer Science

2020

Grossmont College, Associate of Science in Mathematics / Computer Science

2017